

Homework 3

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (3C-1). Suppose $T \in \mathcal{L}(V, W)$. Show that w.r.t each choice of basis of V and W , the matrix of T has at least $\dim \operatorname{im} T$ nonzero entries.

Proof. Pick two basis in V and W , denoted by $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_m\}$ respectively. Set $M := \mathcal{M}(T; \mathcal{B}, \mathcal{C})$ being the matrix of T , and $d := \dim \operatorname{im} T$.

Suppose that M has at most $d - 1$ nonzero entries. Then M has at least $n - d + 1$ columns with all-zero entries, which indicates that there are at least $n - d + 1$ zero vectors among the n vectors $Mb_1, \dots, Mb_n$. Then

$$\dim \operatorname{im} T = \dim \operatorname{span} \{Mb_i : 1 \leq i \leq n\} \leq n - (n - d + 1) = d - 1.$$

Hence, we get a contradiction, and the matrix M has at least d nonzero entries. $\square$

Problem 2 (3C-2). Suppose $T \in \mathcal{L}(V, W)$, where V and W are finite-dimensional and nonzero. Prove that $\dim \operatorname{im} T = 1$ if and only if there exist a basis of V and a basis of W such that, with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1.

Proof. (a) We will prove " $\Rightarrow$ " first. Since $\dim \operatorname{im} T = 1$, there exists a nonzero vector $w_0 \in W$ such that $\operatorname{im} T = \operatorname{span} \{w_0\}$. We can extend w_0 to a basis of W , denoted by $\mathcal{C}' = \{w_0, w_1, \dots, w_{m-1}\}$. Let $w_m = w_0 - w_1 - \dots - w_{m-1}$, it's easy to verify that $\mathcal{C} := \{w_1, \dots, w_m\}$ is also a basis of W .

Next, pick a basis of V in $V - \ker T$ (this is a n -dimensional vector space minus an at most $(n - 1)$ -dimensional subspace, in which there do exist a basis of V), denoted by $\mathcal{B}' = \{v'_1, \dots, v'_n\}$. Since $0 \neq Tv'_i \in \operatorname{im} T = \operatorname{span} \{w_0\}$, there exists a scalar a_i such that $Tv'_i = a_i w_0$. Then set

$$\mathcal{B} := \{a_1^{-1}v'_1, \dots, a_n^{-1}v'_n\}.$$

It's easy to verify that $\mathcal{B}$ is also a basis of V , and for every $v \in \mathcal{B}$, $Tv = w_0 = w_1 + \dots + w_m$. Hence w.r.t the bases $\mathcal{B}$ and $\mathcal{C}$, the matrix of T has all entries equal to 1.

(b) We will prove " $\Leftarrow$ " now. Suppose that there exist a basis of V and a basis of W such that, with respect to these bases, all entries of $\mathcal{M}(T)$ equal 1. Then for every vector $v \in V$, we can write

$v = \sum_{i=1}^n a_i v_i$, where $v_1, \dots, v_n$ are the basis of V . Then

$$Tv = T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T v_i = \sum_{i=1}^n a_i (w_1 + \dots + w_m) = (a_1 + \dots + a_n)(w_1 + \dots + w_m),$$

where $\{w_i : 1 \leq i \leq m\}$ is the selected basis of W . Hence, $\text{im } T \subseteq \text{span}\{w_1 + \dots + w_m\}$, which indicates that $\dim \text{im } T \leq 1$. Since $T \neq 0$, we have $\dim \text{im } T = 1$. $\square$

Problem 3 (3C-6). Suppose $v_1, \dots, v_m$ is a basis of V and W is finite-dimensional. Suppose $T \in \mathcal{L}(V, W)$. Prove that there exists a basis $w_1, \dots, w_n$ of W such that all entries in the first column of $\mathcal{M}(T)$ [with respect to the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$] are 0 except for possibly a 1 in the first row, first column.

Proof. (1) If $Tv_1 = 0$, then we can pick an arbitrary basis of W and the conclusion holds.

(2) If $Tv_1 \neq 0$, we can extend Tv_1 to a basis of W , denoted by $w_1 = Tv_1, w_2, \dots, w_n$. Then w.r.t the bases $v_1, \dots, v_m$ and $w_1, \dots, w_n$, the matrix of T has all entries in the first column equal to 0 except for a 1 in the first row, first column. $\square$

Problem 4 (3C-9). Suppose $a = (a_1 \ \dots \ a_n)$ is a 1-by- n matrix and B is an n -by- p matrix. Prove that

$$aB = a_1 B_{1,\cdot} + \dots + a_n B_{n,\cdot}.$$

In other words, show that aB is a linear combination of the rows of B , with the scalars that multiply the rows coming from a .

Proof. Let $B_{i,\cdot}$ be the i -th row of B . Then by the definition of block matrix multiplication, we have

$$aB = (a_1 \ \dots \ a_n) \begin{pmatrix} B_{1,\cdot} \\ \vdots \\ B_{n,\cdot} \end{pmatrix} = \sum_{i=1}^n a_i B_{i,\cdot}.$$

$\square$

Problem 5 (3C-12). Prove that matrix multiplication is associative. In other words, suppose A , B , and C are matrices whose sizes are such that $(AB)C$ makes sense. Explain why $A(BC)$ makes sense and prove that

$$(AB)C = A(BC).$$

Proof. Suppose $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times k}$. Since $(AB)C$ make sense, $C \in \mathbb{F}^{k \times l}$ for some positive integer l . Then BC is an $n \times l$ matrix, and $A(BC)$ makes sense.

Denote the entries of A , B , and C by A_{ij} , B_{ij} , and C_{ij} respectively. Then

$$((AB)C)_{s,t} = \sum_{j=1}^k \left(\sum_{i=1}^n A_{si} B_{ij} \right) C_{jt}$$

and

$$(A(BC))_{s,t} = \sum_{i=1}^n A_{si} \left(\sum_{j=1}^k B_{ij} C_{jt} \right).$$

By the associative property of scalar multiplication and addition, these two expressions are equal. Thus, $(AB)C = A(BC)$. $\square$

Problem 6 (3C-16). Suppose A is an m -by- n matrix with $A \neq 0$. Prove that the rank of A is 1 if and only if there exist $(c_1, \dots, c_m) \in \mathbb{F}^m$ and $(d_1, \dots, d_n) \in \mathbb{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, \dots, m$ and every $k = 1, \dots, n$.

Proof. (a) We will prove " $\Rightarrow$ " first. Since the rank of A is 1, all the column vectors of A (i.e. $A_{\cdot,i}$ for $1 \leq i \leq n$) lies in a one-dimensional subspace of $\mathbb{F}^m$. Hence, there exists a nonzero vector $c = (c_1, \dots, c_m) \in \mathbb{F}^m$ such that for every $1 \leq i \leq n$, there exists a scalar d_i such that $A_{\cdot,i} = d_i c$. Then we have $A_{j,k} = c_j d_k$ for every $j = 1, \dots, m$ and every $k = 1, \dots, n$.

(b) We will prove " $\Leftarrow$ " now. Suppose there exist $(c_1, \dots, c_m) \in \mathbb{F}^m$ and $(d_1, \dots, d_n) \in \mathbb{F}^n$ such that $A_{j,k} = c_j d_k$ for every $j = 1, \dots, m$ and every $k = 1, \dots, n$. Denote $c = (c_1, \dots, c_m)$. Then, by the definition of rank, we have

$$\text{rank } A = \dim \text{span}\{A_{\cdot,1}, \dots, A_{\cdot,n}\} = \dim \text{span}\{d_1 c, \dots, d_n c\} = 1.$$

$\square$

Problem 7 (3C-17). Suppose $T \in \mathcal{L}(V)$, and $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V . Prove that the following are equivalent:

- (a) T is injective.
- (b) The columns of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{n,1}$.
- (c) The columns of $\mathcal{M}(T)$ span $\mathbb{F}^{n,1}$.
- (d) The rows of $\mathcal{M}(T)$ span $\mathbb{F}^{1,n}$.
- (e) The rows of $\mathcal{M}(T)$ are linearly independent in $\mathbb{F}^{1,n}$.

Here $\mathcal{M}(T)$ means $\mathcal{M}(T, (u_1, \dots, u_n), (v_1, \dots, v_n))$.

Proof. Denote $M := \mathcal{M}(T) = (c_1, \dots, c_n) = (r_1^T, \dots, r_n^T)^T$, where $c_i \in \mathbb{F}^{n \times 1}$ and $r_i \in \mathbb{F}^{1 \times n}$.

(a) $\Rightarrow$ (b): if $c_1, \dots, c_n$ are linearly dependent, then there exists not-all-zero scalars $a_1, \dots, a_n$ such that $a_1 c_1 + \dots + a_n c_n = 0$, which means $T(a_1 u_1 + \dots + a_n u_n) = 0$, contradicting to (a).

(b) $\Rightarrow$ (c): $c_1, \dots, c_n \in \mathbb{F}^{n \times 1}$ are linearly independent, then $\{c_1, \dots, c_n\}$ is a basis of $\mathbb{F}^{n \times 1}$, which implies the conclusion.

(c) $\Rightarrow$ (a): (c) means that T is surjective, hence $V = \text{im } T = \text{span}\{Tu_1, \dots, Tu_n\}$. Since $\dim V = n$, $\{Tu_1, \dots, Tu_n\}$ is a basis of V . Hence, T is injective.

(c) $\Leftrightarrow$ (d): (c) $\Leftrightarrow$ Column rank of M equals $n \Leftrightarrow$ Row rank of M equals $n \Leftrightarrow$ (d).

(d) $\Leftrightarrow$ (e): (d) $\Leftrightarrow \dim \text{span}\{r_1, \dots, r_n\} = n$. Since $\dim \mathbb{F}^{1 \times n} = n$, that is equivalent to $r_1, \dots, r_n$ are linearly independent. $\square$